REPOMIND

Repo-scale coding agent on AMD MI300X

256K context · FP8 · MIT · self-hosted

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AMD Developer Hackathon 2026

Track 1 (AI Agents) + Hugging Face Special Prize + Build-in-Public + Qwen partner

The problem

Closed coding agents leave money — and security — on the table

	Cursor	Claude Code	Copilot	REPOMIND
Open source	❌	❌	❌	✅ **MIT**
Self-hosted on your hardware	❌	❌	❌	✅
Loads **whole** repo, not fragments	❌	partial	❌	✅ **256K**
Per-developer cost	$40/mo	$100/mo	$39/mo	**GPU only**
Banks / defense / pharma allowed	❌	❌	❌	✅

Banks can't use Cursor. Defense contractors can't. Pharma can't. Apple iOS team can't. Their codebases are the proprietary edge — they cannot legally leave their VPC.

The architectural moat

192 GB on a single chip

For Qwen3-Coder-Next-FP8 + 256K context window, single-GPU memory budget:

```
Weights (FP8)                 ~80 GB
KV cache @ FP8 for 256K       ~38 GB
Activations + framework       ~25 GB
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TOTAL                        ~143 GB
```

	NVIDIA H100 80GB	AMD MI300X 192GB
Single-card capacity	80 GB	192 GB
Fits 143 GB workload	❌ requires 2-4× sharding	✅ headroom
Per-card AllReduce overhead	yes	none

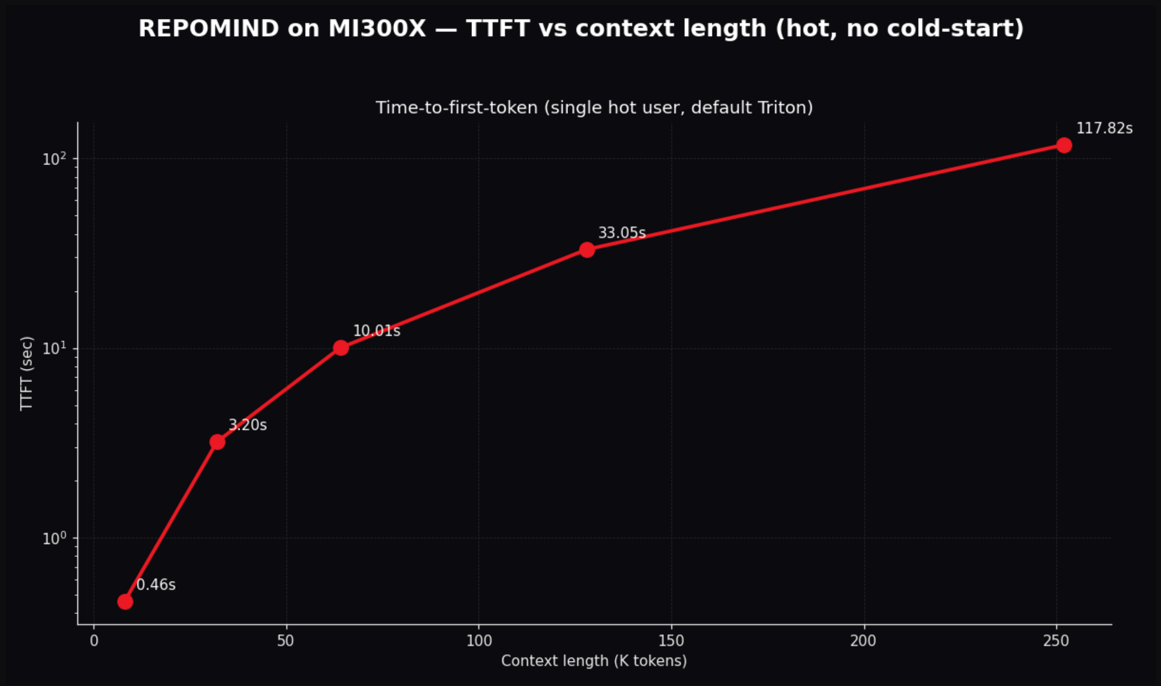
Verified on real hardware — 2026-05-05 / 06

It works. Here are the numbers.

Single MI300X x1 · vLLM 0.17.1 + ROCm 7.2 · 124 min total across 2 sessions · $4.12

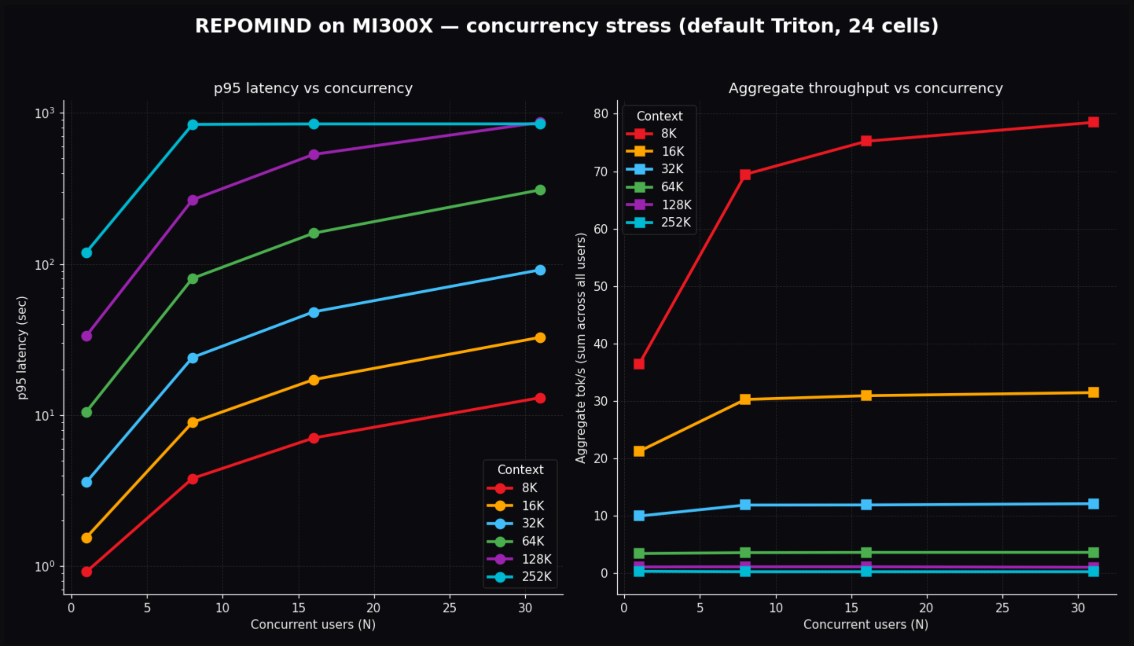
Metric	Verified
Model weights in VRAM	**77.29 GiB**
Available KV cache memory	**94.58 GiB** (2,065,744 tokens)
VRAM peak at full load	**176 / 191.7 GiB** (92% utilization)
`--max-model-len 262144`	started clean, `Application startup complete`
`/v1/models` returns	`max_model_len: 262144` ✅
Cold start (download + load + compile + warmup)	~3 min 30 sec
Warm restart	~1 min 30 sec
[illegible]	[illegible]

Throughput vs context length — hot path, 6 contexts



Context	Prompt	TTFT (hot)	Decode wall	Tok/s	Source
8K	8,090	0.46s	0.94s	agg 78.5 @ N=31	extended
16K	16,224	1.55s	1.55s	agg 31.4 @ N=31	extended
32K	[illegible]	[illegible]	[illegible]	[illegible] (single user)	session 1
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Concurrency stress — 24 cells across 6 contexts



Context	N=1	N=8	N=16	N=31 (success)	Source
8K	36.5	69.4	75.2	**78.5 (31/31 ✅)**	extended
16K	21.2	30.2	30.9	**31.4 (31/31 ✅)**	extended
32K	9.95	11.85	11.87	**12.08 (31/31 ✅)**	session 1
64K	3.41	3.57	3.60	**3.61 (31/31 ✅)**	extended
128K	1.07	1.10	1.10	1.01 (25/31, 6 timeouts)	session 1

Tuning attempt: AITER backend → measured regression

We tried the obvious lever. Here's what we found.

`--attention-backend ROCM_AITER_FA` (AMD's hand-tuned MI300X kernels)

Outcome	Default Triton	AITER (FP8 KV cache)
Output quality (144 cells)	**0/144 broken** ✅	**137/144 broken** ✕
8K × 31 throughput	78.5 agg tps	168 agg tps (+114%)
64K × 31 throughput	3.61 agg tps	18.5 agg tps (+411%)
TTFT @ 64K hot	10.01s	3.54s (2.8× faster)
Sample output	"`longest_common_subsequence` is in `/utils.py`..."	"!!!!!!!!!!!!!!!!!!!!!!!!!"

Long-context coherence — proven at 200K

256K window is **usable**, not just **allocated**

A unique sentinel function `calc_repomind_token_budget_v7` and magic constant `4242` embedded in a ~200K-token code corpus. Model is asked to recover both via JSON.

Position	Prompt tokens	Found function	Found constant	PASS
early	99,814	✅	✅	✅
middle	**199,413**	✅	✅	✅
late	99,814	✅	✅	✅

Many "256K window" claims in the industry are allocated memory only — accuracy collapses past ~64K. **Qwen3-Coder-Next on MI300X actually attends to deep**

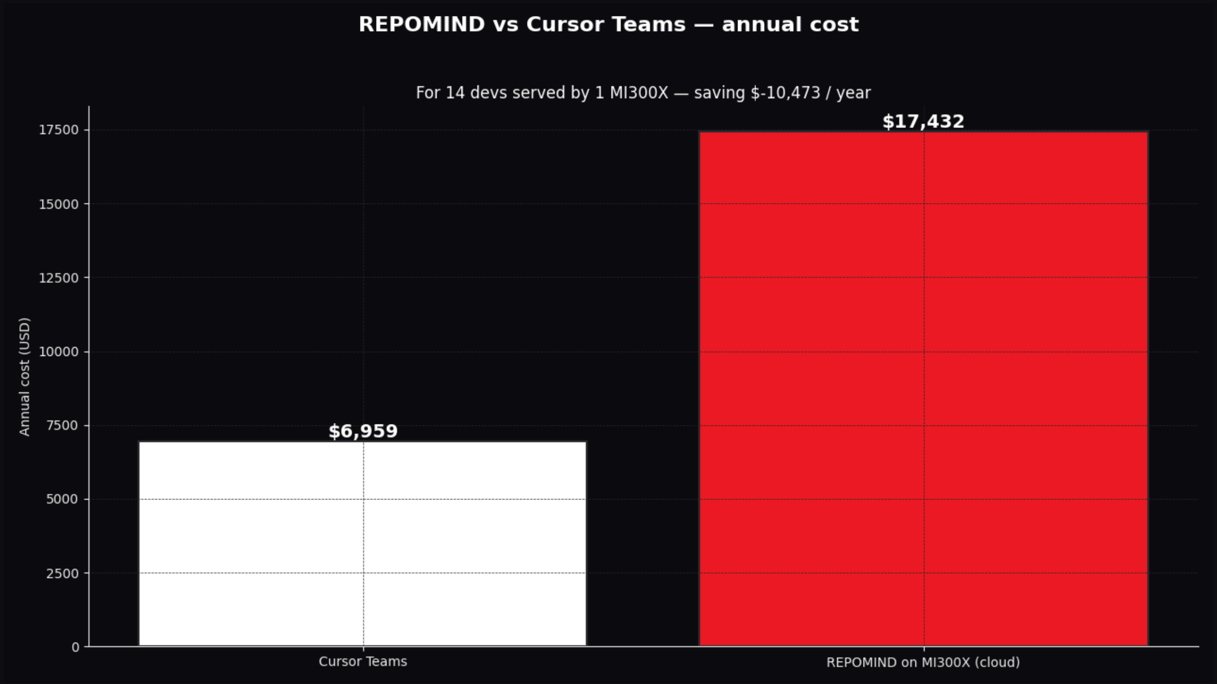
End-to-end repo Q&A — 9/9 correct

Including a repository 5× the context window

Tier	Repo	Repo tokens	Files	Chunks	Q1	Q2	Q3
small	this repo	67,618	68	348	✅	✅	✅
medium	`pallets/flask`	408,447	227	1,995	✅	✅	✅
large	`pytorch/vision`	1,307,491	581	6,799	✅	✅	✅

> Q (pytorch/vision): "Where does video decoding live?"
> A: **"Video decoding lives in the `torchvision.io` module, specifically in `torchvision/io/video.py` and `torchvision/io/video_reader.cpp`. The implementation uses `pyav` (FFmpeg bindings) as the backend..."**

Cost economics



Metric	Value
AMD Cloud rate	$1.99 / GPU / hour
$ per 1M completion tokens (32K, N=31)	$45.75
Active continuous queriers / MI300X	14.5
Bursty engineering team seats / MI300X	70-140

Lisa Su said: "AI is for everyone."

> AMD CEO Lisa Su, CES 2026 keynote, Jan 5

We took it literally. **REPOMIND opens a $5-10B market** that closed coding agents abandoned:

Hyperscalers · Banks · Defense · Pharma · Healthcare · Consumer Tech

Every team that can't or won't ship code to a SaaS coding agent.

AMD made the hardware. We made the open-source MIT unlock.